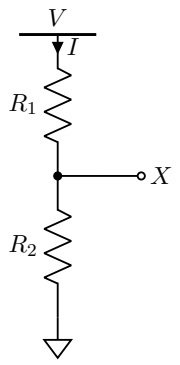
Partitori di tensione e di corrente, potenza ed energia

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Partitore di tensione (a vuoto)



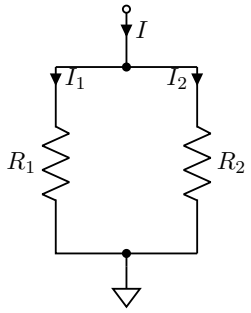
$$I = \frac{V}{R_1 + R_2}$$

$$V_X = V \cdot \frac{R_2}{R_1 + R_2}$$

$$V_1 = V \cdot \frac{R_1}{R_1 + R_2}$$

$$V_1 + V_X = V$$

Partitore di corrente



$$I_1 = I \cdot \frac{R_2}{R_1 + R_2}$$

$$I_2 = I \cdot \frac{R_1}{R_1 + R_2}$$

$$I_1 + I_2 = I$$

$$R_{eq} = \frac{R_1 \cdot R_2}{R_1 + R_2}$$

Potenziometro

al posto di R_2 (verso massa):

$$0 \le V_X \le V \cdot \frac{R_{2\,max}}{R_1 + R_{2\,max}}$$

al posto di R_1 (verso V):

$$V \cdot \frac{R_2}{R_{1\,max} + R_2} \le V_X \le V$$

$$I_{max} = \frac{V}{R_{\text{fissa}}}$$

Potenza

$$P = V \cdot I$$

$$P = R \cdot I^2 \qquad P = \frac{V^2}{R}$$

P [W] V [V] I [A] R [Ω]

resistori: $\frac{1}{4}\,\text{W} = 0{,}25\,\text{W}$ $\frac{1}{2}\,\text{W}$ $1\,\text{W}$

Energia

$$E = P \cdot t$$

E [Wh] P [W] t [h]

$$1\,\text{kWh} = 1000\,\text{Wh}$$

$$1\,\text{Wh} = 3600\,\text{J} \qquad 1\,\text{J} = 1\,\text{W} \cdot 1\,\text{s}$$

costo = energia [kWh] · prezzo [euro/kWh]

Tempo in ore

min	10	15	30
h	$\approx 0{,}167$	0,25	0,5

$t\,[\text{h}] = t\,[\text{min}]/60$

Multipli

$1\,\text{kW} = 1000\,\text{W}$ $1\,\text{mW} = 0{,}001\,\text{W}$

$1\,\text{k}\Omega = 1000\,\Omega$ $1\,\text{mA} = 0{,}001\,\text{A}$